

Kenneth Loi

Doudna Lab, UC Berkeley | Email: kenjmloi@berkeley.edu

EDUCATION

Ph.D., Biophysics

2026–present

University of California, Berkeley

B.A., Molecular and Cell Biology

2024

University of California, Berkeley

RESEARCH INTERESTS

RNA-guided systems; phage defense; biomolecular evolution; CRISPR biology; structural bioinformatics; AI for biology

RESEARCH EXPERIENCE

Graduate Researcher

2026–present

Doudna Lab, UC Berkeley

Previously undergraduate researcher and research technician

2023–2026

- Co-first author of the VIPR discovery study, accepted in *Science*; identified a phage-encoded RNA-guided DNA recognition system with a noncontiguous targeting code.
- Leveraged genomic language models and small RNA-sequencing to discover a novel guide RNA for VIPR.
- Developed structural bioinformatics and comparative genomics pipelines to identify divergent RNA-guided systems and infer their evolutionary relationships.
- Designed and executed plasmid-based functional assays for programmable repression, phage-defense genetics, and guide-target specificity.

Undergraduate Researcher

2021–2023

Murthy Lab, UC Berkeley

- Tested lipid nanoparticle (LNP) formulations for nucleic acid delivery.
- Investigated mRNA chemical modifications to enhance transcript stability and translation efficiency.
- Collaborated on projects delivering gene editing effectors to target tissues.

Summer Undergraduate Researcher (SR-EIP)

2021

Jones Lab, Brown University

- Helped develop the Human Neocortical Neurosolver (HNN): a software for biophysical modeling of MEG/EEG potentials of neural circuits
- Leveraged HNN to study possible mechanisms driving Alzheimer's-associated MEG/EEG readings

PUBLICATIONS

- Yoon PH, **Loi KJ**, Zhang Z, Docter TA, Lopez SC, Langeberg CJ, Moez ur-Rehman M, Vohra K, Zhou Z, Shi H, Boger R, Wang PY, Adler BA, Brohawn SG, Doudna JA. A Noncontiguous Code for RNA-Guided DNA Recognition at the Origin of CRISPR-Cas. *Science*, accepted. **Co-first author**.
- Yoon PH, Docter TA, Zhang Z, **Loi KJ**, Valentin-Alvarado LE, Brohawn SG, Doudna JA. Triplex formation drives noncontiguous VIPR RNA-guided DNA recognition. *bioRxiv*, 2026. doi: 10.64898/2026.04.26.720927.
- Tuck OT, Hu JJ, Lopez SC, Adler BA, O'Brien CE, Hsieh K, Meredith C, **Loi K**, Yoon PH, Doherty EE, Lahiri A, Doudna JA. Recurrent acquisition of nuclease-protease pairs in antiviral immunity. *Science* 391(6781):195–201, 2026. doi: 10.1126/science.aea8769.
- Doherty EE, Adler BA, Yoon PH, Hsieh K, **Loi K**, Armbruster EG, Lahiri A, Bolling CS, Wilcox XE, Akkati A, et al. A miniature CRISPR–Cas10 enzyme confers immunity by inhibitory signalling. *Nature* 647(8091):997–1004, 2025. doi: 10.1038/s41586-025-09569-9.

- Skopintsev P, Esain-Garcia I, DeTurk EC, Yoon PH, Zhou Z, Weiss T, Kamalu M, Chamraj A, **Loi KJ**, Langeberg CJ, et al. Structure and evolution-guided design of minimal RNA-guided nucleases. *Science* 393(6808):313–318, 2026. doi: 10.1126/science.aed6123.
- Yoon PH, Zhang Z, **Loi KJ**, Adler BA, Lahiri A, Vohra K, Shi H, Rabelo DB, Trinidad M, Boger RS, et al. Structure-guided discovery of ancestral CRISPR-Cas13 ribonucleases. *Science* 385(6708):538–543, 2024.
- Yoon PH, Skopintsev P, Shi H, Chen L, Adler BA, Al-Shimary M, Craig RJ, **Loi KJ**, DeTurk EC, Li Z, et al. Eukaryotic RNA-guided endonucleases evolved from a unique clade of bacterial enzymes. *Nucleic Acids Research* 51(22):12414–12427, 2023.
- Jas M, Thorpe R, Tolley N, Bailey C, Brandt S, Caldwell B, Cheng H, Daniels D, Fernandez-Pujol C, Khalil M, **Loi K**, et al. HNN-core: A Python software for cellular and circuit-level interpretation of human MEG/EEG. *Journal of Open Source Software* 8(92):5848, 2023.
- Sharma R, William BE, **Loi K**, Murthy N. Efficient delivery of nucleic acids by lipid nanoparticles to the liver, spleen, and brain. *Molecular Therapy* 31(4):286, 2023.

PRESENTATIONS & POSTERS

A noncontiguous code at the origin of CRISPR	2026
<i>CRISPR 2026 — Talk</i>	<i>June</i>
Scalable Structural Homology Pipeline Reveals Hidden Protein Relationships	2025
<i>Cold Spring Harbor Genome Engineering Meeting — Poster</i>	
Structure-guided discovery of ancestral Cas13s	2024
<i>Sternberg Lab Seminar; Columbia University — Talk</i>	
Biophysical modeling of Alzheimer’s disease mediated changes in MEG alpha oscillations	2021
<i>Annual Biomedical Research Conference for Minority Students — Poster</i>	

AWARDS & FELLOWSHIPS

NSF Graduate Research Fellowship Program	2026
<i>Honorable Mention</i>	
UC Berkeley Rose Hills Fellow	2022
The Leadership Alliance Summer Research - Early Identification Program	2021

TEACHING & MENTORSHIP

SPLASH: Molecular Matrix: Weaving Biology and Machine Learning Together	2024
<i>Taught high school students about how advanced uses of ML in biology</i>	
SPLASH: Gene editing for human health. A CRISPR view and beyond	2022
<i>Taught high school students how CRISPR is impacting human health</i>	

TECHNICAL SKILLS

Wet-lab: Molecular cloning, protein purification, mammalian cell culture (transfection, maintenance), plate-reader assays.

Bioinformatics: HMMER, IQ-TREE, AlphaFold, Foldseek

Programming: Python, Bash, LaTeX.